S6 Text.

EvoSim: connection count and wiring strength under fluctuation (exploratory)

Supporting information for "Environmental fluctuation limits gene regulation unless it can be tracked"

This study began with EvoSim, an individual-based model in which the genome is a regulatory network of evolvable size (S1 Text). Its earlier results are summarised in S4 Text. This experiment measured the strength of regulatory connections as well as their number. It is exploratory. The pre-registered confirmation could not be run (section 4), and EvoSim organisms did not use the cue in this regime, so the experiment concerns fluctuation that cannot be tracked. The main text reports the R-Aevol experiments.

1. Model

EvoSim is an individual-based model of unicellular organisms on a two-dimensional lattice with diffusing, cross-fed resources (S1 Text). Each genome is a gene regulatory network, specified by a weight matrix and node biases, with 13 to 32 nodes. Thirteen nodes are fixed. Twelve are read out as phenotypic traits (resource-uptake allocation, body size, motility, division threshold, DNA repair, stress tolerance and leakage) and one receives an abiotic cue, much like the sensor gene of Sakamoto and Innan [1]. The remaining hidden nodes are added by duplication and removed by deletion, so genome size can evolve. The network is developed once at birth, and the phenotype does not change afterwards. Mutation can create and remove connections and each non-zero connection has a small maintenance cost, so connectivity is also free to evolve.

The design follows the R-Aevol experiments with an abiotic cue. Fluctuating arms have an informative cue, a yoked cue decoupled from the environmental state by running on an incommensurate period, or no cue. In a frozen environment the cue is zero, so the three frozen arms are identical.

EvoSim is described in full, following the ODD protocol [2], in S1 Text; the parts relevant to this experiment are summarised below. Organisms live on a $G \times G$ lattice ($G = 128$) with six diffusing resource channels. External inflow, which varies seasonally, enters the first channel (with the exception described under Environment and cue), and consuming channel r releases a byproduct into channel $r + 1$, with yields declining along this chain. All arms also experience rare random mass-mortality and famine events, on average one every 25,000 time steps. Each organism has an energy store and divides when the store exceeds a heritable threshold. Starvation and the other causes of death are described in S1 Text.

Genome and development. A genome is a weight matrix W and a bias vector b over n_g nodes, with $13 \le n_g \le 32$. Nodes 1 to 12 are read out as the phenotype, node 13 receives the abiotic cue, and hidden nodes $14, \ldots, n_g$ have no fixed meaning. Development takes place once, at birth. Expression is initialised at 0.1 and updated eight times as

$$e_k \leftarrow \tanh\Big(b_k + \sum_m W_{mk}\, e_m + u_k\Big), \tag{1}$$

where u_k is the local concentration of resource k for $k \le 6$, the cue for $k = 13$, and zero otherwise. The final expression values are mapped to the twelve traits.

Mutation. At each birth, a Poisson number of point mutations with mean $10^{-3} n_g^2$ is applied, each adding $\mathcal{N}(0, 0.25^2)$ to a randomly chosen weight. This rate is reduced by up to 85% by the organism's heritable repair trait. With probability equal to the reduced per-weight rate times n_g, one node bias receives a mutation of the same size. The structural mutations are gene silencing (probability 0.002), which sets a node's outgoing weights to zero; gene duplication (0.002), which copies the regulatory role of one node onto another and adds $\mathcal{N}(0, 0.05^2)$ noise to every copied weight; and the addition or deletion of a hidden node (0.003 each), where an added node is a duplicate of an existing one made in the same way. Connection silencing (probability 0.2 per birth) sets one randomly chosen weight to exactly zero. Without

it, additive mutation would drive density towards one. Each hidden node costs 0.0015 and each non-zero weight 5×10^{-5} energy units per time step.

Environment and cue. An abiotic condition cycles through discrete states with a period of 2,000 time steps. Each state changes which resource channel gives the highest yield, and in fluctuating arms 40% of the external inflow goes to the favoured channel. In frozen arms all inflow enters channel 1, which then also gives the highest yield. The phase of the cycle, scaled to $[-1, 1]$, is the input to node 13 and identifies the current state. Fluctuating arms have six states and frozen arms one. In the yoked arm the cue is instead the phase multiplied by $10/\varphi \approx 6.18$, with φ the golden ratio, and wrapped back into the same range, so that it takes the same values but carries only 0.0004 bits of information about the state. In the no-cue arm the gain of the cue is zero. With a single state the cue is identically zero, so the three frozen arms produce identical runs and were analysed as one arm.

Measures. For every organism at birth we record the number of weights with $|w| > 10^{-4}$, the strength density $\sum |w|/n_g^2$ over all n_g^2 weights, including zeros, and n_g. The log holds their population means at each time step. Count density is the mean number of non-zero weights divided by the squared mean n_g, and total strength is the mean strength density multiplied by the squared mean n_g. Each measure is then averaged over the last 20% of the run. Populations that went extinct were excluded and their number is reported.

Runs. We ran six arms with 30 seeds each for 150,000 time steps (180 runs), about 6,200 generations in fluctuating and 4,900 in frozen populations (24 and 31 time steps per generation, from the slope of mean lineage depth). Runs are deterministic for a given seed and, on the same machine and Julia version, bit-identical across thread counts (checked at 1, 4 and 8 threads).

2. Results

Earlier EvoSim experiments had found that fluctuation increases connection density (S4 Text). We repeated the comparison in 180 runs (six arms of 30 seeds, 150,000 time steps each) and measured the strength of connections as well as their number. The cue is zero in a frozen environment, so the three frozen arms are identical and the runs contain 120 distinct populations. Without a cue, fluctuation raised count density, the proportion of possible connections that are non-zero, from 0.467 to 0.587 (+25.5%, $p < 10^{-4}$, $d = 2.2$; 28 surviving fluctuating and 30 frozen populations; Fig A, panel A), and the number of non-zero connections from 86 to 120 (+39%, $p < 10^{-4}$). Total wiring strength did not change (38.0 in frozen and 37.5 in fluctuating populations, $p = 0.86$; Fig A, panel C). Fluctuating networks thus had more connections but not more wiring, and the average non-zero connection was about 30% weaker. Strength density, the summed absolute weight divided by the number of possible connections, fell from 0.207 to 0.184 (−10.9%, $p < 10^{-4}$, $d = -1.3$; Fig A, panel B), mostly because there were more possible connections. Node count stayed close to its minimum of 13 in this regime and increased only slightly (+0.6 nodes, $p = 0.047$), which raises n_g^2 by about 9%. With the informative and the yoked cue, fluctuation had effects in the same directions (count density +18% and +28%, total strength unchanged, $p = 0.40$ and 0.59).

Population size is a possible confound. Fluctuating populations were 40% smaller than frozen ones, as expected since in frozen dishes all inflow feeds the best-paying channel (S1 Text), and population size can change genome architecture by itself. Because the population sizes of the two arms did not overlap, the two effects cannot be separated statistically, and the correlations within arms point in different directions for the two measures. Larger populations had lower strength density in every arm (r between −0.31 and −0.47; Fig A, panel D), so population size would predict higher strength density in the smaller fluctuating populations, the reverse of what we found. Count density, in contrast, was higher in smaller populations within each fluctuating arm (r between −0.63 and −0.72), though not in the frozen arm ($r = 0.24$), so population size may contribute to the increase in connection number. The cue was also not used in this

regime. Populations with the informative and the yoked cue were the same size ($p = 0.79$) and had the same strength density ($p = 0.99$), although count density was lower with the informative cue (0.55 against 0.60, $p = 0.009$). EvoSim organisms respond to the cue only when development is reversible, and even then they match their phenotype to the current state only in part (S4 Text). The EvoSim results concern fluctuation that the organisms cannot track.

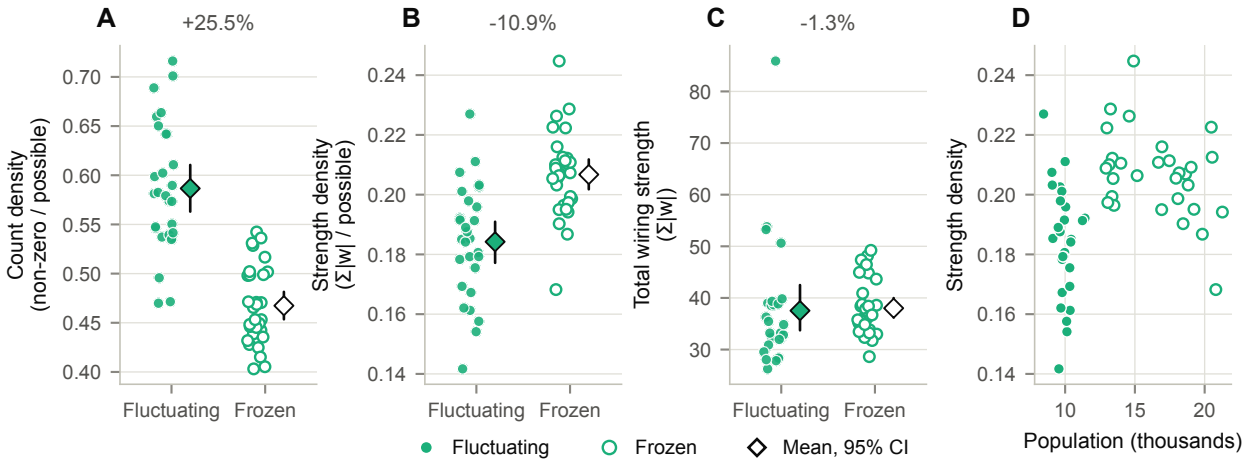


Fig A. EvoSim: more connections, no more wiring. Fluctuating (no cue; filled) versus frozen (hollow) populations, one point per population; diamonds are means with bootstrap 95% confidence intervals. Numbers above panels are the change in the mean. (A) Count density, the number of non-zero connections divided by the number of possible connections. (B) Strength density, the summed absolute weight divided by the number of possible connections. (C) Total wiring strength. (D) Strength density against population size within each arm. Extinct populations (2 of 30 fluctuating, 0 of 30 frozen) are excluded.

3. Interpretation

Gene duplication and node addition copy the connections of an existing node and add small random values to every copied weight, including weights that were zero, so each duplication can create many weak connections. Point mutations that hit a zero weight also create connections. If purifying selection is weaker in the smaller populations of fluctuating environments, such connections would persist and connection number would rise without an increase in total strength, which fits the correlations with population size within the fluctuating arms. We have not tested this mechanism.

4. Limitations and the confirmation that was not run

Fluctuation and population size are confounded between arms. The within-arm correlations argue against population size as the cause of the lower strength density but are consistent with a contribution to the higher connection count. We pre-registered a confirmation of the divergence between count and strength (P5; commit `4bc16c3`) in a regime with more hidden nodes, on condition that fluctuating and frozen populations survived a viability pilot there. They did not (2 to 5 of 6 fluctuating populations survived at grids 80, 96 and 112, compared with 6 of 6 frozen populations; commit `56fe3a0`), so the confirmation was not run and the result remains exploratory.

References

1. Sakamoto T, Innan H. On the evolutionary origin of discrete phenotypic plasticity. G3 (Bethesda). 2024;14(9):jkae144.
2. Grimm V, Railsback SF, Vincenot CE, Berger U, Gallagher C, DeAngelis DL, et al. The ODD protocol for describing agent-based and other simulation models: a second update to improve clarity, replication, and structural realism. J Artif Soc Soc Simul. 2020;23(2):7.